

Literature-implied partial answer to a point-finite ball covering question

Source arXiv:1212.2817; supporting arXiv:2007.05242

Source question

Fonf, Levin, and Zanco, *Covering L^p spaces by balls* (arXiv:1212.2817), ask in Question 1.1:

Which infinite-dimensional Banach spaces admit point-finite coverings by balls, each of positive radius?

Their main theorem gives a negative answer for separable infinite-dimensional Banach spaces that are both uniformly rotund and uniformly smooth; in particular, for separable infinite-dimensional $L^p(\mu)$, $1 < p < \infty$.

Supporting theorem

De Bernardi, *A note on point-finite coverings by balls* (arXiv:2007.05242), explicitly cites the Fonf–Levin–Zanco paper and states that Klee’s problem about point-finite coverings by balls of $\ell_p(\Gamma)$, $1 < p < \infty$, remained open in the nonseparable case. Theorem 3.6 of that paper proves the following strengthened negative result: if X is infinite-dimensional, uniformly rotund and uniformly smooth, and

$$\text{dens}(X) < 2^{\aleph_0},$$

then X admits no point-finite covering by open or closed balls, each of positive radius. The same theorem also proves that any cover of such an X by open balls is not point-finite.

Status

This is a literature-implied partial answer to Question 1.1 of arXiv:1212.2817. It extends the source paper’s separable uniformly rotund/smooth obstruction to all density-below-continuum uniformly rotund/smooth spaces. It does not solve the full classification problem: arXiv:2007.05242 explicitly leaves open whether the density restriction can be removed, even for Hilbert spaces.

Search evidence

The local run indexes had no existing packet for arXiv:1212.2817. Web search for the exact source question and for “point-finite coverings by balls” returned arXiv:2007.05242 as the direct later development, and the parsed source of arXiv:2007.05242 contains the explicit citations and theorem stated above.

References

- V. P. Fonf, M. Levin, and C. Zanco, *Covering L^p spaces by balls*, arXiv:1212.2817.
C. A. De Bernardi, *A note on point-finite coverings by balls*, arXiv:2007.05242.